

# Private Markets SWES — Stress Testing Framework for Private Credit Funds

Applying the Bank of England Private Markets System-Wide Exploratory Scenario via FIS APT with fundamental credit and liquidity overlays

Prepared for: Senior Risk Manager, Alternatives (Private Credit) | Date: 10 July 2026 | Status: Framework v1.0 — for internal methodology sign-off

Scenario source: Bank of England, *Private markets SWES: publication of the stress scenario* (scenario phase launched 2026; exercise announced Dec 2025), incl. Variable Paths annex (5-year paths, 190+ variables)

## 1. Executive Summary

**The exercise.** The BoE private markets SWES is a five-year, severe-but-plausible global stagflationary recession — a supply/geopolitical shock with rising energy prices, high inflation, **rising policy rates** (Bank Rate averaging 7% in Year 2), a deep output contraction, sharply tighter financial conditions, impaired exit markets, and a deliberately slow recovery. Severity is calibrated to the Bank Capital Stress Test (BCST). It is a two-round behavioural exercise across 46 participants: the Bank wants your *losses and your actions* (valuation behaviour, fund extensions, amendments, financing responses), not just a P&L number.

**Answer 1 — Can the annex shocks be applied directly to APT?** Largely yes, and this is the key design insight: **the BoE has already performed the economic-to-financial scenario expansion** that we used to do internally in bank stress tests. The Variable Paths annex specifies not only macro variables but the full set of market risk factors: corporate bond spreads by rating notch (AAA–CCC) in GBP and USD, leveraged loan price *and* nominal spread indices (US and European), a AAA CLO price index, CDS indices, complete government and swap curves (GBP/USD/EUR), equity indices with sector splits, EV/EBITDA multiples by sector, VIX, and sector revenue paths. These map onto APT factors with only unit/timing conversion. Residual expansion is needed for exactly three things: **(i)** private credit / direct-lending discount margins (not a published index in the annex — must be expanded from the leveraged loan and single-B spread paths), **(ii)** FX (no currency paths are provided), and **(iii)** any APT factor granularity the annex does not reach (e.g. sector-level credit spreads).

**Answer 2 — Is the floating-rate-note (FRN) setup fine?** Yes — modelling each loan as an FRN (reference rate + contractual margin, reset frequency, expected repayment date, current mark) is the correct and standard vehicle in a factor model, *provided* exposures are assigned analytically (spread duration × margin shock) rather than estimated by regression on smoothed private NAVs, which would understate risk materially. Under this scenario the rate-duration term is nearly irrelevant (floaters reset), so P&L is driven almost entirely by **discount-margin widening × spread duration**. That is also the trap: a pure factor-model run will show a manageable mark-to-market loss while missing the real damage — reference rates near 7% plus widened margins push weak borrowers' interest coverage below 1× just as sector revenues fall ~10–15% cumulatively. Hence the two-engine architecture below.

**Answer 3 — Architecture.** Run **two engines plus a liquidity/behavioural lens**, reconciled: **Engine 1** (FIS APT, top-down): factor shocks → instantaneous MTM at Year-1 and peak-stress (end-Year-2) points. **Engine 2** (bottom-up fundamental): sector revenue and EV/EBITDA paths → borrower EBITDA, ICR and LTV under stressed base rates → rating migration, defaults, recoveries → impairment overlay and revised marks. The annex hands you the inputs for Engine 2 deliberately — sector revenues and EV/EBITDA multiples exist in the file precisely so private-asset holders can revalue bottom-up. **Lens 3:** fund-level liquidity (undrawn commitments, subscription/NAV lines, refinancing walls in the impaired Years 1–3 exit market, hedge margin) and the behavioural actions the SWES templates request.

### Bottom line

Apply the annex market-variable paths directly to APT after unit/timing conversion (Section 5). Expand only the three gap variables using the satellite-beta methodology in Section 7. Model loans as FRNs with analytically assigned spread durations. Add the fundamental migration/default overlay (Section 6) — under this scenario it, not the MTM, is where the loss story lives — and reconcile to avoid double counting (Section 8).

## 2. The Scenario at a Glance

Narrative: a global supply and geopolitical shock disrupts supply chains (tech hardware acutely), spikes energy (Brent averages \$125 in Y1, \$155 in Y2), and produces stagflation: UK CPI averages 7% in Y1 while real GDP contracts ~2% in each of Y1 and Y2. Policy tightens into the recession — Bank Rate and Fed funds average 6% (Y1) and 7% (Y2). Financial conditions tighten sharply; recovery is deliberately slow (weak productivity), with markets recovering slower than past crises. Corporate revenues fall while funding costs rise and exit markets are impaired — a pincer designed to stress leveraged capital structures.

| Variable (annex definition)                           | Y1            | Y2            | Y3          | Peak / trough        | Modelling relevance                             |
|-------------------------------------------------------|---------------|---------------|-------------|----------------------|-------------------------------------------------|
| UK real GDP (annual change)                           | -2.0%         | -2.0%         | +0.7%       | ≈ -4% trough         | Engine 2 demand anchor                          |
| US real GDP (annual change)                           | -1.5%         | -1.5%         | +0.7%       | ≈ -3% trough         | US borrower books                               |
| UK CPI (annual change)                                | +7.0%         | +4.5%         | +2.7%       | 7% peak              | Stagflation — no rate relief                    |
| Bank Rate / Fed funds (annual avg level)              | 6.0%          | 7.0%          | 6.0%        | 7% avg Y2            | Coupon ↑, borrower ICR ↓                        |
| SONIA / SOFR overnight (annual avg level)             | 6.0%          | 7.0%          | 6.0%        | 7%                   | FRN reference rate path                         |
| UK unemployment (annual avg level)                    | 5.2%          | 6.4%          | 7.5%        | 7.5% (Y3)            | Consumer-facing sectors                         |
| GBP HY corporate spread (cum. change)                 | +390bp        | +790bp        | +590bp      | +790bp end-Y2        | APT HY factor shock                             |
| GBP single-B spread (cum. change)                     | +440bp        | +860bp        | +650bp      | +860bp end-Y2        | Closest rating proxy for direct lending         |
| USD single-B spread (cum. change)                     | +530bp        | +950bp        | +740bp      | +950bp end-Y2        | US mid-market loans                             |
| USD CCC spread (cum. change)                          | +600bp        | +1,320bp      | +960bp      | +1,320bp end-Y2      | Watch-list / stressed names                     |
| GBP / USD IG spread (cum. change)                     | +120/130bp    | +230/240bp    | +180/190bp  | +230–240bp           | IG sleeves, ABF                                 |
| Morningstar LSTA US Lev Loan nominal spread (cum.)    | +200bp        | +400bp        | +300bp      | +400bp end-Y2        | <b>Primary anchor for PC expansion</b>          |
| Morningstar LSTA US Lev Loan price index (annual chg) | -28%          | -10%          | +11%        | ≈ -35% trough        | Cross-check on FRN revaluation                  |
| European Lev Loan spread / price                      | +200bp / -28% | +400bp / -10% | —           | +400bp / ≈ -35%      | EUR loan books                                  |
| J.P. Morgan AAA CLO price index (annual chg)          | -13%          | -5%           | +5%         | ≈ -17% trough        | CLO tranches; NAV-line collateral               |
| iTraxx XOVER / CDX HY 5Y CDS (cum. change)            | +470/420bp    | +870/820bp    | —           | +870bp end-Y2        | Hedge legs, if any                              |
| FTSE All-Share / S&P 500 (annual chg)                 | -31% / -36%   | -6% / -6%     | +11/12%     | ≈ -35% / -40%        | Equity co-invest; VIX avg 40 in Y1              |
| UK / US EV/EBITDA multiple, aggregate (annual chg)    | -26% / -32%   | +4%           | +4%         | -26% / -32% Y1       | Engine 2: LTV & equity cushion                  |
| Sector revenues, UK/US aggregate (annual chg)         | -5%           | -5%           | +4%         | ≈ -10% cum.          | Engine 2: EBITDA paths (sector splits in annex) |
| UK / US CRE prices (annual chg)                       | -13% / -16%   | -21% / -25%   | —           | ≈ -31% / -37% trough | RE-backed lending collateral                    |
| UK 10y gilt / US 10y UST (annual avg level)           | 5.9% / 6.0%   | 5.6% / 5.7%   | 4.8% / 4.9% | ≈ 6% Y1              | Curve factors; inverted vs Bank Rate Y2         |

Source: BoE Private Markets SWES Variable Paths annex (all 193 rows retained in the accompanying workbook *SWES\_variable\_paths\_BoE\_annex.xlsx*). “Cum. change” rows are the sum of annual changes from jump-off; “annual avg level” rows are levels as published. Sector-level paths (11 GICS-style sectors × UK/US for revenues, equities and EV/EBITDA; 9 rating notches × 2 currencies for spreads) are in the annex.

#### What is deliberately painful for private credit in this calibration

- **Stagflation, not GFC:** rates rise ~300bp and stay high — floating-rate coupons increase (carry offset in the fund P&L) but borrower cash interest costs roughly double. At SOFR  $\approx 7\%$  + 550bp margin, cost of debt  $\approx 12.5\%$ ; a 5× levered borrower with EBITDA down 10–15% sees ICR fall towards or below 1×.
- **Slow recovery by design:** spreads peak at end-Y2 and only partially retrace by Y5; exit/refinancing markets stay impaired — 2027–29 maturity walls cannot assume friendly refis. Fund-extension and amendment behaviour is exactly what the Bank wants to observe.
- **Valuation opacity is a named vulnerability:** the Bank is explicitly probing whether private marks lag public comparables. Submissions whose private credit marks move materially less than the LSTA –28%/–35% path will need a defensible bridge.

### 3. Design Decision: Direct Application vs Scenario Expansion

In bank stress tests (Fed CCAR, PRA ACS/BCST, HKMA SPM) the regulator supplies a macro scenario and firms expand it to market risk factors via satellite models, because macro variables alone barely move a trading book. The SWES inverts this: because participants are market-facing non-banks, the Bank pre-expanded the scenario into financial variable paths. The decision framework is therefore:

| Bucket                                           | Variables                                                                                                                                                                                                                                         | Treatment                                                                                                                                                                            |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A. Direct to APT</b><br>(conversion only)     | Swap and government curves (GBP/USD/EUR, all tenors); corporate spreads by rating × currency; leveraged loan spread and price indices; CLO AAA price; CDS indices; equity index and sector paths; VIX; EV/EBITDA multiples                        | Convert units and timing (Section 5.3); map to APT factor set; apply as factor-level shocks. No modelling discretion beyond conventions.                                             |
| <b>B. Expansion required</b> (satellite mapping) | <b>Private credit / direct-lending discount margins</b> (unitranche, senior MM, junior); <b>FX</b> (no paths in annex); sector- or strategy-level spread tilts (e.g. software vs consumer direct lending); NAV-financing advance-rate assumptions | Satellite regression / stress-beta on annex anchors (Section 7), documented and sign-off'd. Check the participant guidance pack first — it may prescribe conventions (e.g. FX flat). |
| <b>C. Not factor-model territory</b>             | Sector revenue paths, unemployment, GDP, house prices, CRE (as collateral fundamentals)                                                                                                                                                           | Engine 2 inputs: borrower fundamentals, migration, default and recovery (Section 6); collateral haircuts for asset-backed positions.                                                 |

#### Practical note — check the submission pack

The BoE states that participants received *detailed guidance and templates* alongside the scenario. Before finalising conventions, confirm in the pack: the jump-off date; whether FX is to be held flat; whether marks are to be reported at year-end points or annual averages; and the required granularity (fund / strategy / position). Where the pack prescribes a convention, it overrides the defaults in this framework.

### 4. Target Architecture

Three components, one reconciliation:

|                 | Engine 1 — Market (FIS APT)                                                            | Engine 2 — Fundamental credit                                                                                    | Lens 3 — Liquidity & behaviour                                                     |
|-----------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Question</b> | What is the instantaneous MTM of the portfolio if the annex market shocks materialise? | Which borrowers migrate, default, or breach covenants over the 5-year path, and what impairment does that imply? | Can each fund meet its obligations through Years 1–3, and what actions do we take? |
| <b>Inputs</b>   | FRN-proxied loan terms; APT factor shocks from bucket A + expanded bucket B            | Borrower financials; sector revenue paths; stressed base rates; EV/EBITDA paths; migration matrix                | Undrawn commitments, sub/NAV lines, maturity ladder, hedge CSAs, LP profile        |
| <b>Horizon</b>  | Instantaneous: Y1 point and peak (end-Y2) point; optional Y5 recovery point            | Annual, Years 1–5                                                                                                | Quarterly, Years 1–3                                                               |
| <b>Output</b>   | MTM Δ by fund/strategy/factor; spread DV01; concentrations                             | Migration table, default/loss schedule, stressed marks, watch-list                                               | Liquidity coverage, actions log (the SWES behavioural submission)                  |

**Why both engines:** spread widening in Engine 1 prices the *market's* expectation of loss on a still-performing loan; Engine 2 captures the *realised* credit deterioration (non-accruals, PIK toggles, restructurings) that dominates private credit outcomes in a five-year recession. Either alone under- or double-counts; Section 8 reconciles.

## 5. Engine 1 — FIS APT Implementation

### 5.1 Position representation: the FRN proxy

Represent each loan with the following minimum fields. This is the standard vendor-model treatment and is adequate for factor P&L; the loan-specific credit story lives in Engine 2.

| Field                                                         | Convention                                                                                                                                                       | Why it matters                                                                                               |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Reference rate + tenor                                        | SONIA / SOFR / Euribor per facility; actual reset frequency (typ. 1–3m)                                                                                          | Drives the (small) rate-duration leg; coupon carry in multi-period view                                      |
| Contractual margin                                            | Current cash margin; PIK component flagged separately                                                                                                            | Carry; PIK flags are Engine 2 early-warning names                                                            |
| Rate floor                                                    | Capture, but note: with reference rates at 6–7% floors are far out-of-the-money in this scenario — no floor optionality P&L                                      | Avoid spurious floor valuation effects                                                                       |
| Maturity: <b>expected repayment date</b> , not legal maturity | Use expected-repayment / weighted-average-life; under stress, <i>extend</i> expected repayment (prepayments stop, amend-and-extend) — typically +1y vs base case | Sets spread duration. Under stress WAL lengthens exactly when spreads widen — a real, often-missed amplifier |
| Current mark / discount margin (DM)                           | Solve DM from current fair value; this is the state variable to be shocked                                                                                       | Anchors revaluation to your actual marks                                                                     |
| Rating (internal), sector, currency, seniority                | Internal rating mapped to public scale (typically B/B– for sponsor-backed direct lending); GICS-style sector matching annex sectors                              | Selects which annex spread path and which Engine 2 revenue path applies                                      |

### 5.2 Valuation mechanics

$\Delta V/V \approx -SD \times \Delta DM - RD \times \Delta r_{ref} + t \times (r_{ref} + \text{margin})$  [carry term for multi-period views]  
 where  $SD$  = spread duration  $\approx$  duration to expected repayment (typ. 2.0–3.5y for MM direct lending)  
 $RD$  = rate duration  $\approx$  time to next coupon reset (typ. 0.1–0.25y) — near zero for floaters

**Order of magnitude:** a +300bp DM shock (Section 7 expansion of the +400bp leveraged loan peak) on  $SD \approx 2.75$  gives  $\approx -8\%$  MTM before defaults; the annex LSTA price path (–28% Y1,  $\approx -35\%$  trough) is B/CCC-heavy and includes distress/liquidity components, so a diversified performing direct-lending book marking –8% to –15% at peak, *plus* the Engine 2 impairment overlay, is the defensible corridor. Use the annex price-vs-spread pair as an internal consistency check: price implied by (spread path  $\times$  index duration) vs published price path — the residual is the default/liquidity component that Engine 2 must supply.

#### Critical APT pitfall — do not let regression betas near private marks

APT estimates factor exposures by regression on return histories. Private credit NAV series are appraisal-smoothed and stale; regression on them yields artificially low betas and  $R^2$ , and the model will report false diversification and understated stress loss. **Assign exposures analytically:** set up positions as FRN terms & conditions so APT prices them cash-flow-analytically (spread duration route above), or override betas with de-smoothed values. If APT insists on returns-based estimation for any sleeve, de-smooth first (Geltner unsmoothing:  $r_t^* = (r_t - \phi \cdot r_{t-1}) / (1 - \phi)$ ,  $\phi = AR(1)$  coefficient of the NAV series, typically 0.4–0.6 for private credit) and validate the resulting beta against  $\approx 0.6$ –0.9 on the leveraged loan index.

### 5.3 Converting annex paths into APT factor shocks

The annex mixes two forms; convert both to *instantaneous shocks vs jump-off* for the factor model:

| Annex form                           | Variables                                                                                       | Conversion to factor shock                                                                                                                                                                                                                                                  |
|--------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| “Annual change” ( $\Delta$ per year) | Spreads, equity, EV/EBITDA, loan/CLO price indices, revenues, GDP, property                     | Cumulative sum from jump-off: $\text{shock}(Y1) = \Delta_1$ ; $\text{shock}(\text{peak}) = \Delta_1 + \Delta_2$ . For % variables compound: $(1+\Delta_1)(1+\Delta_2)-1$ .                                                                                                  |
| “Annual average level”               | All rates/curves (Bank Rate, SONIA/SOFR/Euribor swaps, gilt/UST/Bund yields), unemployment, VIX | Shock = stressed level – jump-off level. For end-of-year points on a monotone path, interpolate: $\text{level}(\text{end-Y1}) \approx (\text{avg } Y1 + \text{avg } Y2)/2$ . For the peak run use the peak annual average (Y2) directly – mildly conservative, document it. |

**Recommended run set** (each an instantaneous APT scenario):

1. **Run A – “Year 1”**: cumulative Y1 changes; end-Y1 interpolated rate levels. The acute repricing.
2. **Run B – “Peak stress”**: cumulative Y1+Y2 changes; Y2 average rate levels. The binding MTM number; spreads peak at end-Y2 across the annex.
3. **Run C – “Y5 residual”**: cumulative Y1–Y5 changes. Quantifies the deliberately incomplete recovery (e.g. GBP HY still +230bp vs jump-off; useful for “slow recovery” narrative and vintage/exit analysis).
4. **Sensitivities**: Run B with (i) PC expansion beta at upper bound, (ii) WAL +1y, (iii) FX stress leg if unhedged sleeves exist.

#### 5.4 SWES-variable → APT factor mapping (bucket A)

| SWES annex variable                 | APT factor family                          | Shock (Run B, peak)                                                                          | Notes                                                       |
|-------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| SONIA swap curve (o/n–20y)          | GBP swap curve nodes (key rates)           | To Y2 avg levels, e.g. o/n → 7.0%, 5y → 5.5%, 10y → 4.8%                                     | Curve inverts — apply node-by-node, don't parallel-shift    |
| SOFR swap curve                     | USD swap curve nodes                       | o/n → 7.0%, 5y → 5.9%, 10y → 5.1%                                                            | Same treatment                                              |
| Euribor swap curve                  | EUR swap curve nodes                       | o/n → 5.0%, 5y → 3.8%, 10y → 3.3%                                                            | EUR shocked less than GBP/USD                               |
| Gilt / UST / Bund zero curves       | Government curve factors                   | Per-tenor levels from annex                                                                  | Drives any bond sleeves, LDI-style overlays                 |
| GBP/USD corporate spreads by rating | Credit spread factors by rating × currency | IG +230/240bp; BB +680/750bp; B +860/950bp; CCC +760/1,320bp                                 | Rating-notch granularity — use borrower internal-rating map |
| Lev loan nominal spreads (US/EU)    | Loan/HY spread factor (or custom factor)   | +400bp both                                                                                  | <b>Anchor for PC expansion (Section 7)</b>                  |
| CLO AAA price                       | Securitised/CLO factor                     | ≈ –17% price trough                                                                          | Also informs NAV-line collateral haircuts                   |
| iTraxx / CDX CDS                    | CDS spread factors                         | XOVER +870bp; CDX HY +820bp; IG +210bp                                                       | Hedge legs; basis vs cash from annex pairs                  |
| Equity indices + sectors            | Equity market/industry factors             | FTSE ≈ –35%, S&P ≈ –40% trough; sector tilts per annex (RE, tech, financials –41 to –44% Y1) | Equity co-invest and warrant positions                      |
| VIX                                 | Volatility factor                          | Avg 40 in Y1                                                                                 | Any optionality; convexity sleeves                          |
| EV/EBITDA multiples by sector       | (Not an APT factor)                        | UK –26% / US –32% Y1                                                                         | Engine 2 input: LTV and equity-cushion revaluation          |

## 6. Engine 2 — Fundamental Credit Overlay

This is the engine the SWES design is really aimed at, and it reuses your existing credit review machinery with scenario-driven inputs. Annual steps, Years 1–5, per borrower (top-30 names individually; remainder by sector × rating cohort):

1. **Revenue path:** apply the annex sector revenue path (UK or US, 11 sectors — e.g. consumer discretionary –10%/–11% Y1/Y2, energy +8%/+7%) to each borrower's revenue. For UK borrowers in sectors without a direct match, use the aggregate (–5%/–5%).
2. **EBITDA path:** apply operating leverage:  $\Delta \text{EBITDA}\% = \Delta \text{Revenue}\% \times \text{OL}$ , with OL from historical borrower data or a default 1.5–2.5× by sector; add an inflation-margin squeeze for Y1 (CPI 7% vs pricing power) for consumer-facing names.
3. **Cash interest:** recompute at stressed reference rates (SONIA/SOFR path: 6% Y1, 7% Y2) + contractual margin. This is where the stagflationary design bites.
4. **Credit metrics:**  $\text{ICR} = \text{EBITDA}_s / \text{cash interest}_s$ ; leverage = net debt/EBITDA<sub>s</sub>; LTV = net debt/(EBITDA<sub>s</sub> × stressed EV/EBITDA multiple). The multiple path (–26/–32% Y1) collapses equity cushions even where EBITDA holds.
5. **Migration:** map metrics to internal rating via your masterscale; alternatively apply a stressed migration matrix (Moody's/S&P B-rated corporate matrices scaled to recession intensity — GFC-vintage matrices are the right severity comparator). Flag covenant breaches (typically  $\text{ICR} < 1.5\times$ , leverage headroom < 10%).
6. **Default & loss:** PD from migrated rating (through-the-cycle B default rates roughly 3–4%/yr, rising to 8–12%/yr in recession vintages; CCC 20%+); LGD for senior secured MM loans 30–40% baseline, stressed to 45–55% given EV multiple compression and impaired exits; ECL = PD × LGD × EAD per year, cumulative over Y1–Y3.
7. **Stressed marks:** re-mark each surviving loan at DM = expanded scenario DM (Section 7) + migration add-on (e.g. +150–250bp per notch downgrade below B); defaulted names to recovery value. Compare fund-level stressed NAV to Engine 1 — the bridge is your reconciliation (Section 8) and your answer to the Bank's valuation-opacity probe.

### Behavioural items the templates will ask for (prepare narratives with numbers)

Amend-and-extend vs enforce decisions; PIK toggling policy; reserve/valuation-committee marking policy in a dislocated market; use of continuation vehicles and fund extensions; new-deployment appetite at wide spreads (dry powder is countercyclical capacity — a genuine offset worth stating); LP communication and any gating/redemption mechanics for semi-liquid vehicles; demands on and from financing counterparties (sub-line and NAV-line pricing/availability).

## 7. Scenario Expansion Methodology (Bucket B) — Replicable Recipe

For any target variable X absent from the annex, the methodology is the satellite-model approach used in CCAR/ACS internal expansions, kept deliberately simple and auditable:

1. **Select anchors  $A_1 \dots A_k$**  from annex variables economically linked to X (max 2–3; multicollinearity kills stability).
2. **Estimate** on the longest clean history (monthly/quarterly,  $\geq 10y$ , must span GFC and/or COVID):

$$\Delta X_t = \alpha + \sum \beta_i \cdot \Delta A_{i,t} + \varepsilon_t \quad (\text{OLS, Newey-West s.e.})$$

Cross-check with a **stress beta**:  $\beta^{\text{stress}} = \Delta X^{\text{crisis}} / \Delta A^{\text{crisis}}$  measured peak-to-peak over GFC and COVID windows — if regression and stress betas diverge, prefer the stress beta (regressions on calm data understate tail co-movement).

3. **Apply** betas to annex paths → path for X. Add a prudence margin where fit is poor:  $X^{\text{final}} = X^\beta + \lambda \cdot \sigma(\varepsilon)$ ,  $\lambda = 0.5$  (default) or 1.0 (weak fit,  $R^2 < 0.5$ ).
4. **Document** anchors, sample, coefficients,  $R^2$ , crisis back-check, and margin; sensitivity at  $\beta \pm 1$  s.e.; sign-off via model/methodology governance. Refresh annually or on scenario change.



## 7.1 Worked expansion — private credit discount margin (the key gap)

| Step                 | Specification                                                                                                                                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Target X             | $\Delta DM$ on senior middle-market direct lending (repeat for unitranche and junior/mezz with separate calibrations)                                                                                                                                                   |
| Anchors              | $A_1$ = Morningstar LSTA leveraged loan nominal spread (annex: +200/+400bp); $A_2$ = USD single-B corporate spread (annex: +530/+950bp)                                                                                                                                 |
| Calibration data     | Cliffwater Direct Lending Index (CDLI) three-year-takeout yields; Lincoln International Senior Debt Index fair-value marks; KBRA DLD spread data; your own fund marks history (de-smoothed). Quarterly, 2008– or 2015– depending on series.                             |
| Expected calibration | Historically private credit DMs widen with $\beta \approx 0.6$ –0.9 to broadly syndicated loans, with a 1–2 quarter reporting lag and less retracement in recovery. <b>Working assumption pending your regression: <math>\beta = 0.75</math> on the LL spread path.</b> |
| Resulting path       | $\Delta DM_{PC} \approx +150bp$ (Y1), <b>+300bp (peak, end-Y2)</b> , +225bp (Y3), $\approx +75bp$ residual (Y5); + prudence margin per step 3; + Engine 2 migration add-ons per name (these come on top and differentiate the book)                                     |
| Sensitivity band     | $\beta = 0.6 \rightarrow +240bp$ peak; $\beta = 1.0 \rightarrow +400bp$ peak (i.e. “private = BSL, no smoothing benefit” — a sensitivity the Bank will like)                                                                                                            |

## 7.2 Other bucket-B expansions

| Gap variable                              | Recipe                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FX (GBP/USD, EUR/USD, EUR/GBP)            | First check the guidance pack — likely “hold constant”. If own-view needed: risk-off USD appreciation via stress beta of DXY/GBPUSD to VIX over GFC/COVID (order of magnitude: GBP –10–15% vs USD). For hedged share classes the P&L item is <b>hedge roll cost</b> at the stressed forward-point differential (annex gives all three money-market curves — compute directly) plus margin/collateral usage in Lens 3. |
| Sector spread tilts within private credit | Annex sector <i>equity</i> and <i>revenue</i> dispersion as the tilt driver: scale the PC DM shock by sector-revenue beta (e.g. consumer discretionary 1.3×, healthcare 0.8×, energy 0.5×). Keep portfolio-weighted average equal to the 7.1 path (tilt, don't add).                                                                                                                                                  |
| NAV financing / fund-finance facilities   | Not a market factor: model as Lens 3 item — advance-rate tightening (haircut widening consistent with CLO AAA –17% and LL –35% collateral paths), spread re-pricing at renewal (+150–300bp consistent with IG/bank funding paths), and availability risk in Y1–Y2.                                                                                                                                                    |
| Real-asset / infra debt sleeves (if any)  | Anchor to IG/BBB spread paths + CRE price path for collateral; infra revenues typically low-beta to the sector revenue paths (use utilities path: +2/+3% — mildly positive).                                                                                                                                                                                                                                          |



## 8. Reconciliation and Double-Count Control

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Engine 1's DM widening already embeds the market's pricing of expected loss; Engine 2 computes expected loss directly. Adding them naively double-counts. Convention (standard in IFRS 9-meets-MTM stress frameworks):

```
Scenario loss (peak) = MTME1(performing book, scenario DM) + [Recovery value - prior  
mark]defaulted names, E2  
+ Migration add-on MTM (E2 notch-downgrades priced at annex rating-spread differentials)  
- Carry offset (coupon income at stressed reference rates × average performing balance, if  
multi-period view requested)
```

- Names that **default** in E2 are removed from the E1 MTM population from the year of default and carried at recovery value — no name is both spread-widened and written down in the same period.
- The migration add-on uses the *differences* between annex rating-notch spread paths (e.g. B→CCC in USD ≈ +370bp incremental at peak), so it is additive to the generic PC DM path without overlap.
- Sanity anchors: fund-level stressed NAV change should sit between the CLO AAA path (−17%, senior/diversified floor) and the LSTA loan price trough (−35%, B/CCC-heavy ceiling) depending on book seniority and quality. Present the bridge: generic DM widening → migration add-ons → defaults → carry offset. That bridge is your valuation-opacity answer to the Bank.

## 9. Outputs and Reporting Pack

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1. **Headline table** per fund: NAV impact at Run A / Run B / Run C, split MTM vs impairment vs carry; spread DV01 and WAL pre/post.
2. **Loss bridge** (Section 8) — one chart per fund.
3. **Migration and watch-list table** — rating distribution jump-off vs Y2 vs Y5; covenant-breach and ICR<1× counts; top-10 stressed names with actions.
4. **Liquidity table** — quarterly sources/uses Y1–Y3: undrawn commitments, sub/NAV line capacity and renewal assumptions, hedge margin, refinancing wall.
5. **Behavioural narrative** — the SWES template answers, each anchored to a number from the above.
6. **Methodology annex** — factor mapping, conversion conventions, expansion regressions with diagnostics, de-smoothing parameters, reconciliation rules. This is what survives challenge — from the Bank in Round 1 feedback, and internally from model governance.

## 10. Workplan — Six Weeks, End to End

| Week | Workstream                                                                                                                                                                             | Deliverable                                               |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 1    | Read guidance pack & templates; fix conventions (jump-off, FX, reporting points); position data quality — FRN fields per Section 5.1 for every facility; internal-to-public rating map | Conventions memo; clean position file                     |
| 2    | Bucket A mapping: build the SWES→APT factor shock file (Runs A/B/C); validate APT prices the FRN proxies analytically (no regression betas on private NAVs)                            | APT scenario files; validation note                       |
| 2–3  | Bucket B expansion: run the CDLI/Lincoln/KBRA regressions; fix $\beta$ with sensitivity band; FX/hedge-cost treatment                                                                  | Expansion paper (Section 7 format) → methodology sign-off |
| 3–4  | Engine 1 runs A/B/C + sensitivities; cross-check vs LSTA price path                                                                                                                    | MTM results pack                                          |
| 3–5  | Engine 2: top-30 borrowers individually, cohort model for the tail; migration, ECL, stressed marks                                                                                     | Credit results pack; watch-list                           |
| 5    | Reconciliation bridge; liquidity lens; behavioural narratives with the portfolio teams                                                                                                 | Draft submission tables                                   |
| 6    | Challenge session (CRO/valuation committee); finalise submission + methodology annex; archive for Round 2 re-runs (feedback loop will require updated behavioural responses)           | Final SWES submission; reusable engine for Round 2        |

### Build once, use thrice

Round 2 of the SWES (post-feedback), the July 2026 FSR follow-ups, and your own ICARA/board risk appetite work can all reuse this: the factor-shock file, the expansion regressions and the two-engine reconciliation are scenario-agnostic plumbing. Parameterise the scenario as an input table, not hard-coded numbers.

Framework prepared as an internal methodology document. Scenario data: Bank of England Private Markets SWES Variable Paths annex (accompanying workbook). The scenario is hypothetical and not a forecast. All betas marked “working assumption” require calibration on licensed data (Cliffwater, Lincoln, KBRA, Morningstar/LSTA) before submission use. Verify guidance-pack conventions before finalising.